IWACU RWANDA — Master Project Specification and Delivery Plan

Version 1.0 · 2026-10-27 · Owner: eddyizere24-collab · Author and builder: Claude · Operator (deployment): Grok

0. How to use this document

- This is the single source of truth for **what** is being built, **why**, **how**, **in what order** and **with what proof**. The owner keeps it and uses it to check that every promise is being kept.
- Every claim that something is finished must come with the **evidence** named in section 8 and section 14. Without evidence the status is not DONE.
- Status labels used everywhere: **DONE** (built and proven in a real environment), **BUILT, UNVERIFIED** (written and unit-tested in the author's sandbox, but never built or run in a real environment), **NOT STARTED**, **BLOCKED** (waiting on something outside the builder).
- If this document and a chat message disagree, the owner decides and this document is updated (new version number and date in section 16).
- Related documents: the Collaboration Agreement (Claude × Grok), the Kinyarwanda copy bank (D1), the design canvas (D2 directions and D3 design rules). Links are in section 16.

1. The idea

Problem. A person in Rwanda who travels, or is new to an area, often does not know who leads the village, cell, sector or district they are standing in, and in an emergency may not know whom to call. The information exists, but it is scattered and hard to reach.

Solution. Open the website on any phone and tap one button. The phone's GPS finds the village. The site then shows the chain village → cell → sector → district → province, the leaders of each level with photo and role, **every way to reach them** (call, WhatsApp, e-mail and more), and the **emergency numbers**, which are always shown first and also work without internet.

Purpose. Bring leadership closer to the people. That is why leaders' contact details are public by design.

Principles (never traded away):

1. Emergency first: the emergency numbers are the first thing on screen and need no location.
2. Nationwide from day one: no district is special-cased in code, data or wording.
3. Contact details of leaders are public (owner decision).
4. No citizen accounts, no profiles, no stored location history.
5. Nothing is invented: every public fact has a source and a person who verified it. Boundaries are never made up.
6. Data can be added and changed by non-technical people, without a developer.
7. It must work on cheap Android phones and thin data connections.
8. Honesty: nobody says 'done' without proof, and 'not run' is said when something was not run.

2. Who uses it

- **Citizen:** any age, often a low-end Android phone, limited data, Kinyarwanda first, possibly stressed. Needs the answer in seconds and one big button.
- **Leader:** keeps their own contact details and photo up to date; changes appear at once.
- **Admin (district, sector…):** manages leaders and local emergency numbers for their own area only.
- **Super-admin (owner):** manages roles, contact kinds, national emergency numbers, admins and boundaries for the whole country.
- **Operator (Grok):** deploys, runs the real checks, configures Supabase and Vercel, reports evidence.

3. Decisions frozen by the owner

These are not re-designed by Claude or Grok. A conflict is reported, not resolved silently.

1. The product covers all of Rwanda from the first release.
2. Leaders' contact details are public; a leader may have many contacts (phones, WhatsApp, e-mail, social links) and every one is tappable.
3. Leaders edit their own contacts and photo, and the change is visible immediately (no review). Name and role changes by admins still go back to review.
4. Leader roles and contact kinds are data. An admin adds new ones without a developer.

5. Emergency numbers are shown first and work offline. Each carries its source. Verified now: 112, 111, 113, 3511, 110, 997, 118 (Rwanda National Police site, retrieved 2026-10-26). Hidden until the owner confirms: 114, 116, 3512 (only found in a 2016 article). Not loaded: 912 (sources disagree on what it is).
6. Boundaries come only from an authoritative source with a recorded licence. No placeholder boundary ever ships.
7. No invented (test) leaders in a public deployment until they carry a visible 'Igerageza' mark and their contacts are not tappable.
8. Visual identity: **Direction C, Agaseke** (peace basket with a sun), in the colours of Rwanda used in an original way; light mode by default; one screen with no scrolling; emergency strip fixed at the bottom. Photos of the place are planned for a later version, with a reserved slot now.
9. Copy: the Kinyarwanda copy bank (D1) is approved as written: 'wowe' register, headline 'Menya abayobozi b'aho uri.'
10. The project is the owner's personal project. It may later be offered to the government.

4. What a citizen gets (target product)

C1. **Home screen**, one screen, no scrolling: logo, help '?', language switch, hero place card, one big button 'Menya aho ndi', emergency strip at the bottom with 'Hamagara 112' and 'Izindi nimero'. C2. **Location with consent:** a short explanation before the phone's own permission prompt; clear guidance if GPS is refused or inaccurate. C3. **Result screen:** village, cell, sector, district, province; confidence wording; several candidate villages if unsure; leaders of each level. C4. **Leader card:** photo (or initials), name, role, every contact as a large tappable button (call, WhatsApp, e-mail, social), 'Verified' mark, last-updated date. C5. **Emergency:** national numbers (and local ones for the place) as large buttons; works offline from a copy saved on the phone. C6. **Offline:** the shell and emergency numbers work with no internet; leaders and places need internet until offline sector packs exist. C7. **Help sheet** and **privacy sheet** in plain Kinyarwanda, and Kinyarwanda / English switch. C8. **Errors** for every situation (19 public messages) that say what happened and what to do, never blaming the person. C9. **Later:** photos of the area after the village is found; place-name search; an SMS or USSD channel for people without smartphones (an idea only, not committed).

5. What leaders and admins get (target product)

A1. **Leaders:** list, create, edit, verify, reject, archive (never deleted), with contacts. Four-eyes rule: the creator cannot verify their own entry (except a super-admin). A2. **Settings screens (built, unverified):** emergency numbers (national: super-admin; local: admin of that area; verify needs a recorded source and a second person), leader roles and contact kinds (super-admin). A3. **Bulk import** of leaders from CSV or Excel with a preview and a row-by-row report. A4. **Leader self-service:** a leader signs in and edits their own contacts and photo; 'undo last change'; the unit's admin is alerted when a leader's contacts change. A5. **Photo upload** to our own storage (instead of external links). A6. **Admin users** and their scopes (country → district → sector → cell → village). A7. **Boundary upload** through the validated pipeline, with an integrity report. A8. **Audit:** every change is recorded (who, what, before, after) and cannot be edited or deleted.

6. How it works technically

Stack. Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS; Supabase (PostgreSQL with PostGIS and ltree, Auth); Upstash Redis (rate limiting); Groq (optional place-name router, never sees personal data); Vercel (hosting); GitHub (code and CI); Playwright (end-to-end tests).

Data model (database). `boundaries` (country → province → district → sector → cell → village, each with a unique scope code beginning RW, a parent link, a geometry and a verification status), `data_sources` (source, publisher, version, date, licence), `boundary_history`, `leaders` (status workflow, optional account link), `leader_roles`, `leader_contacts`, `contact_kinds`, `emergency_contacts` (with source and verification), `admin_profiles` (scope), `audit_logs` (append-only).

Public API. POST /api/locate, GET /api/leaders, GET /api/emergency, GET /api/health. **Admin API.** /api/admin/leaders, /api/admin/roles, /api/admin/contact-kinds, /api/admin/emergency; /auth/callback completes the e-mail magic-link sign-in.

Location logic. The database intersects the GPS accuracy circle with verified village polygons, returns a ranked list with a confidence value, then walks up through the explicit parent links. A point outside Rwanda, in a gap, or on an overlap each has a defined answer.

Security and privacy. Row Level Security on every table; the new tables are closed to browsers and served only by the API; writes require same-origin, a signed-in admin, scope checks in the database and in the app, and rate limits; verification needs a second person; every change is audited; links are limited to tel:, sms:, mailto: and https://. Citizens: no

accounts; coordinates are sent to the server to find the village and are not stored; the IP address is used briefly for rate limiting and the hosting provider may log it. Any public statement must say this accurately.

Offline. A service worker precaches the shell and a copy of the verified national emergency numbers; the last good list is also saved on the phone. Offline sector packs are a later step.

Performance targets (to be measured, not yet proven). First screen at most 120 KB in total; usable within 3 seconds on a slow 3G Android; one web font only; illustrations as inline SVG under 6 KB; no carousels or autoplay.

7. Who does what

- **Owner:** decisions; accounts; boundary data and licence; confirming 114, 116, 3512; entering or approving real leader data; consent wording and legal review; testing on real phones; final sign-off.
- **Claude:** all application code, SQL, tests, documentation, copy, design rules, and the versioned project zip with a checksum manifest. Claude cannot reach GitHub, Vercel or Supabase from this chat and works in a sandbox with no network, so it says 'not run' when something was not run.
- **Grok:** replaces the repository from the zip, runs install and all checks, deploys, configures Vercel and Supabase as instructed, runs the SQL, and returns the full evidence in the agreed report. It changes application logic only as a reported last resort.

8. Delivery plan

Two tracks run side by side. Each step lists its goal, who works, what 'done' means and what evidence proves it. Percentages are estimates of overall readiness.

Track A — Engineering stages

Stage 1 — Foundation (to about 40%). Goal: a nationwide, privacy-correct core with leaders, contacts, roles, emergency numbers and tests. Status: **BUILT, UNVERIFIED**. A real Vercel build of a partial copy reached READY (commit d548af8), which is real evidence for the files it contained; three real errors it exposed in Claude's zip were fixed. Evidence still needed: install, typecheck, lint, test, build outputs on the full project.

Stage 2 — First deployment (about 40% → 50%). Goal: the full project runs on a real URL with a real database. Claude: fix every real error found. Grok: apply the latest zip wholesale (verify the checksum manifest), install, commit the lockfile, run the checks, create the Supabase project with postgis and ltree, run `sql/apply_all.sql` on a branch or throw-away project first, set the environment variables (names only in reports), add the redirect URL, explain why /api/health returned HTML, merge to the production branch when green. Owner: accounts and access. Done when: the site opens on its URL, /api/health reports ok, the emergency strip shows the seven verified numbers, the migrations ran with no error. Evidence: command outputs, commit SHA, deployment state, health JSON, SQL results. Status: **BLOCKED** (waiting for Grok's evidence on the full project; at last report the repository was a partial copy, Supabase was not configured and the live /api/health returned HTML).

Stage 3 — Systems to add data without a developer (about 50% → 70%). 3a settings screens for emergency numbers, roles and contact kinds: **BUILT, UNVERIFIED**. 3b bulk import: NOT STARTED. 3c leader self-service, undo, alerts: NOT STARTED. 3d photo upload: NOT STARTED. 3e admin users and scopes: NOT STARTED. 3f boundary upload: NOT STARTED (needs Stage 4 data). Done when: a non-technical admin adds, edits and imports data without touching code or SQL. Evidence: owner's hands-on test and a recording or screenshots.

Stage 4 — National boundaries (about 70% → 82%). Owner obtains the data and licence (National Institute of Statistics / National Land Authority or equivalent) and approves the code table. Claude adapts the ingest tool and checks. Grok loads level by level and reports `boundary_integrity_report()` after each. Done when: all levels are loaded, the integrity report has no error, load and query times are measured. Status: NOT STARTED.

Stage 5 — Leaders and the citizen experience (about 82% → 92%). 'Igerageza' mark for test leaders; consent field and privacy notice; place-name search; offline sector packs; place photos; full Kinyarwanda and English. Owner confirms 114, 116, 3512 and supplies or approves real leader data. Done when: a citizen sees correct leaders for a real village on a real phone. Status: NOT STARTED.

Stage 6 — Release (about 92% → 100%). Close every gate in section 9. Grok: backups with a tested restore, monitoring and alerts, rollback, load test. Owner: legal review, field tests, final sign-off. Status: NOT STARTED.

Track B — Design stages (visual identity and copy)

D1 Copy bank: DONE (approved as written). **D2 Three directions:** DONE (owner chose C, Agaseke). **D3 Design rules:** delivered on the design canvas; awaiting the owner's approval. **D4 Rebuild of the public pages** to the rules: NOT STARTED; one screen with no scrolling at 360×640 and 412×915, emergency strip fixed at the bottom, place card with idle, locating and found states, GPS explanation before the phone prompt, result screen, leader card, help and privacy sheets, all 19 error screens, all copy from D1. Claude checks it at those two sizes before delivering; Grok deploys; the owner reviews on a real phone. **D5 Corrections** after the owner's review (at most two rounds). **D6 Admin screens** restyled with the same rules.

9. Release gates (all must be green, with evidence)

1. Typecheck, lint, unit tests and build pass on the full project; the lockfile is committed.
2. All migrations and the SQL tests run on PostgreSQL; privilege tests as an ordinary signed-in user fail where they should.
3. End-to-end tests pass; the public pages are checked on real Android phones.
4. Only verified leaders and verified emergency numbers are public; every emergency number has a recorded source and the owner has re-confirmed it.
5. No placeholder boundary or invented leader is in the public deployment.
6. Boundaries: integrity report clean; licence recorded.
7. Security: no secrets in the repository; rate limiting active in production; admin writes fail closed without Redis; dependency audit reviewed.
8. Privacy: notice matches reality; legal review of Law No. 058/2021 done; leaders' consent recorded.
9. Accessibility: contrast at least 4.5:1, touch targets at least 48 px, text zoom to 200%, keyboard and screen-reader pass.
10. Performance targets measured and met, or the owner accepts the measured numbers.
11. Backups run and a restore was tested; monitoring and alerts work; rollback rehearsed.
12. The owner signs off.

10. Data governance

- **Boundaries:** official source only; record source, publisher, version, retrieval date and licence; load through the validated pipeline; new geometry stays pending until verified; old versions are kept.
- **Leaders:** created pending; verified by a second person; archived, never deleted; contacts public; a leader's own edits to contacts and photo are immediate and audited.
- **Emergency numbers:** official source recorded on every row; pending until verified; changing the number or wording of a verified row returns it to pending.
- **Photos:** need permission and a source; shown only after the person or admin has supplied them; always with alternative text; small file size.
- **Roles and contact kinds:** switched off, never deleted; codes never change.

11. Quality and testing

- Unit tests for all pure logic (currently 131 passing in the author's sandbox, including tests that read the admin routes to check the security shape).
- SQL tests for the schema, workflows, privileges and constraints (written, never executed).
- End-to-end tests in a browser (written, never executed).
- Real-device tests on low-end Android phones and a slow connection (not done).
- Design checks at 360×640 and 412×915 before each delivery.
- Each delivery ships with a checksum manifest so the repository can be proven identical to the zip.

12. Risks and open decisions

1. Hostile or wrong contact details from a compromised leader login: mitigated by admin-created accounts and audit; 'undo' and alerts are planned (Stage 3c).
2. Test leaders being phoned by citizens: no invented leaders in public until marked (Stage 5).
3. A wrong emergency number: source and verification required; owner re-confirms before launch.
4. Personal data law and consent for publishing contact details: legal review and consent field needed (Stage 5 and 6).

5. Use of national symbols: the sun mark is original and not a copy of the flag or the state emblem; a lawyer is consulted before publication.
6. Boundary data availability and licence: the largest schedule risk; owner-led (Stage 4).
7. Low smartphone ownership and data cost: lightweight design; an SMS or USSD channel is an open idea.
8. Everything built so far is unverified in a real environment; Stage 2 is the first real test and may reveal errors.
9. Open decisions: who verifies local emergency numbers in each district; the consent wording; whether the dark mode is offered; which place-photo source is acceptable.

13. Working rules

1. Claude builds all application code; Grok operates. Neither changes frozen decisions.
2. One source of truth: the versioned zip (`iwacu-rwanda-stageN…zip`) with `MANIFEST.sha256` . The repository must equal it.
3. No green by cheating: no ignored build errors, no disabled lint rules added only to pass.
4. Every delivery lists changed files in the execution log and what was and was not run.
5. Evidence in every report: commit SHA, deployment state, full output of failures, health JSON, environment variable names, SQL results, every edit other than file placement.
6. Claude states limits plainly ('not run', 'not verified'), and corrects earlier mistakes openly.
7. Secrets never appear in chat, commits or logs.
8. Design work needs the owner's approval of each design step before the next is built.

14. Master checklist (owner ticks when evidence is seen)

Foundation

- [ ] Full project builds with typecheck, lint, tests (evidence shown)
- [ ] Migrations 001–003 and seed run without error
- [ ] /api/health returns ok on the production URL
- [ ] Production URL serves the verified code (not an older deployment)

Citizen product

- [ ] Home screen fits one screen at 360×640 and 412×915, no scrolling
- [ ] Emergency strip fixed at the bottom, works offline

- [ ] GPS explanation, consent and refusal guidance as in D1
- [ ] Result screen and leader cards with photo and every contact
- [ ] All 19 error messages as in D1
- [ ] Help and privacy sheets accurate

Admin product

- [ ] Leaders create, edit, verify, archive (four-eyes)
- [ ] Settings: emergency numbers, roles, contact kinds
- [ ] Bulk import, leader self-service with undo, photo upload, admin users, boundary upload

Data

- [ ] Boundaries loaded for the whole country, integrity report clean, licence recorded
- [ ] Real leaders loaded and verified, consent recorded
- [ ] 114, 116, 3512 confirmed by the owner; 912 resolved

Release

- [ ] All gates in section 9 green
- [ ] Legal review done
- [ ] Real-phone field test done
- [ ] Owner sign-off

15. The last dot — what '100%' means

A citizen, in a real village, on a low-end Android phone with a weak connection, opens IWACU RWANDA. Within about three seconds they see the emergency strip and one big button, with no account and no scrolling. They tap it, understand why their location is needed, allow it, and within a few seconds see the correct village, cell, sector, district and province, and the verified leaders with photo and working call, WhatsApp and e-mail buttons. They can call 112 even with no internet. A leader can update their own details and the change is visible at once. An admin with no technical skill can add leaders and local emergency numbers. Every number and boundary on screen has a recorded source and a verifier. The system is backed up, monitored and can be rolled back. A lawyer has reviewed the privacy position, leaders have consented, real phones have been tested, and the owner has signed off.

16. References and change log

- Live site: https://iwacu-rwanda.vercel.app (state to be re-verified by Grok)
- Repository: https://github.com/eddyizere24-collab/Iwacu-rwanda-
- Collaboration Agreement (Claude × Grok): https://claude.ai/artifact/Lu3Mdajksw8DDL7MzQNsNg
- Kinyarwanda copy bank (D1): https://claude.ai/artifact/DwJkdCwPaU4oYfyQHV9yx1
- Design canvas, directions (D2) and design rules (D3): https://claude.ai/artifact/WJc7KC3JTXTSgfEa7KAttd
- Latest project zip at the time of writing: `iwacu-rwanda-stage3a.zip`
- Glossary: Umudugudu = village; Akagari = cell; Umurenge = sector; Akarere = district; Intara = province; Umuyobozi = leader; Ubutabazi = emergency; Hamagara = call.

Change log. v1.0 · 2026-10-27 · first complete specification, written after decisions D1 (copy), D2 (direction C) and the delivery of D3 (design rules).